

HAT-P-12b

R-0802

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_220422 · created 2026-07-28T09:38:38+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459688.6082 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.025
Transit depth (Rp/Rs)^2	2.8 +/- 0.83 [%]
Orbital Inclination (inc)	89.95 +/- 0.89
Airmass coefficient 1 (a1)	1.541 +/- 0.02
Airmass coefficient 2 (a2)	-0.41 +/- 0.26
Scatter in the residuals of the lightcurve fit is	1.31 %
Best Comparison Star	#3 - [116, 303]
Optimal Aperture	2.99
Optimal Annulus	9.66
Transit Duration (day)	0.0524 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.167 $\pm$ 0.025 vs 0.1406 (deviation 0.92 σ)
Duration fit vs published	0.0524 d vs 0.0975 d (deviation 13.04 σ)
Mid-time O-C	-4631.0 min
Depth SNR	5.8
Residual scatter	1.31 %

Light curve

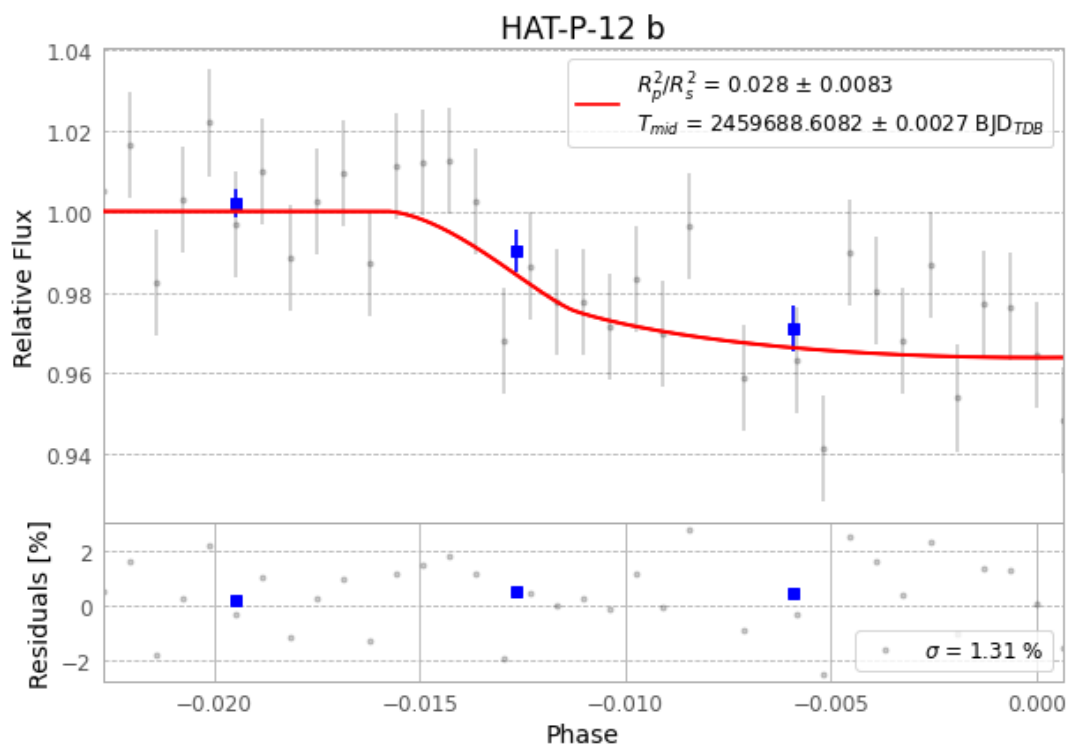


Figure 1 — FinalLightCurve_HAT-P-12 b_2022-04-21.png (SHA-256 652bfc2b559ba447...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [190, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0e8c71ff4d8bec63...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

35 FITS frames (23 MB), HATP-12220422055114.FITS → HATP-12220422073913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-220422.log (1545857846cc...), FinalLightCurve_HAT-P-12 b_2022-04-21.png (652bfc2b559b...), FinalLightCurve_HAT-P-12 b_2022-04-21.csv (228a77881c10...)

Compute resources

Wall time 275.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 09:38Z.